

AKASH TIWARY

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EXPERIENCE

Seven Robotics

July 2023 - Present (3 years 1 months)

Full Stack Developer

- Engineered a production-grade SPA using React 18 and TypeScript, translating Figma designs into pixel-perfect, accessible interfaces, including interactive 3D map visualizations built with ThreeJS, passing design QA review with zero visual defects logged across release cycles.
- Centralized state management using Redux Toolkit and RTK Query with normalized state patterns and optimistic updates, eliminating prop drilling and reducing unnecessary render cycles by 40%.
- Implemented performance and scalability optimizations including React.lazy-based code splitting, route-level chunking, memoization, virtualization, and canvas frame throttling, reducing bundle size by 35% and maintaining 60fps interaction performance.
- Architected and implemented scalable RESTful APIs in Node.js and Express using layered controller-service-repository architecture, improving maintainability, modularity, and separation of concerns across core backend services.
- Built real-time backend services with Socket.IO/WebSockets, WebRTC, and RabbitMQ-driven asynchronous workflows to support live camera feed streaming across multiple robots, task updates, event processing, and reliable background job execution.
- Optimised data layer performance using Redis for caching alongside MySQL, and MongoDB for persistence, improving query efficiency and supporting high-concurrency production workloads.
- Containerized full-stack applications with Docker, creating optimized multi-stage builds and orchestrating local environments with docker-compose
- Deployed a production Node.js server on AWS EC2, configuring a custom VPC with 2 subnets, Internet Gateway, and route tables to manage inbound traffic routing and network isolation.
- Provisioned a dedicated EC2 instance with MongoDB as the database layer, integrating it with the Node.js application server to handle persistent data storage — simulating a self-managed cloud database deployment on AWS.

PROJECTS

Multi-Agent Research Pipeline | Python, Langchain, Langgraph, RAG, Tavily, ChromaDb

[Repo-link](#)

- Built an autonomous AI research agent using LangChain and LangGraph that decomposes queries into sub-questions, runs a Web Search Agent (Tavily) and RAG Agent (ChromaDB) in parallel via LangGraph's fan-out/fan-in pattern, and compiles results into a structured, cited report using Gemini-2.5-flash
- Implemented a hybrid retrieval architecture combining live web search with a persistent vector store (ChromaDB + OpenAI embeddings) supporting PDF, URL, and text ingestion; shipped a Streamlit UI with real-time pipeline progress, knowledge base management, and markdown report export.

Blogging App | React.js, Node.js, Express.js, Firebase, AWS S3, Redis, RabbitMQ

[Repo-link](#)

- Developed a social blogging platform using React.js and Node.js/Express.js with user authentication, image uploads, following system, likes, and commenting features supporting 1000+ user interactions
- Implemented Redis caching for top 100 searched posts (70% faster retrieval), RabbitMQ for push notifications, Firebase/AWS S3 for data/media storage, and secure file handling with Multer and bcrypt authentication

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Node.js

Development: React.js, Docker, Next.js, Tailwind CSS, GraphQL

Databases: MongoDB, Firebase, MySQL, Redis, AWS S3

AI/ML Integration: Langchain, Langgraph, RAG, ChromaDb

Other Techs: AWS, Cloudinary, Drizzle ORM, RabbitMQ, Git